Band-1 Computational Experiment: Lehmer Pairs / Mahler Measure — Next Examples

Human + ZoraASI full-dash (executor)

2026-09-11

Band-1 Computational Experiment: Lehmer / Mahler Measure (Collapse 1.11)

Honesty stamp. This note (i) ledgers the literature on **Lehmer's question** and the smallest known Mahler measures $M > 1$, and (ii) issues **box numerical certificates** for a fixed list of integer polynomials (cyclotomic exclusions $M = 1$; Lehmer polynomial sanity $\tau \approx 1.176280818$; nearby Salem / Pisot checks). It does **not** claim Lehmer's conjecture solved; it does **not** claim a new record below τ; it does **not** elevate informal Zenodo deposits over peer-reviewed / Mossinghoff-aligned tables.

GATE-04: LOCKED · no Clay · no ToE.

Abstract

Mahler measure. For $P = a_n \prod_i (x - \alpha_i) \in \mathbb{Z}[x]$,

$$M(P) = |a_n| \prod_i \max(1, |\alpha_i|).$$

Kronecker: $M(P) = 1$ iff all roots are roots of unity (cyclotomic case, up to units).

Lehmer's question (1933). Does there exist $C > 1$ such that $M(P) > 1 \Rightarrow M(P) \geq C$ for all $P \in \mathbb{Z}[x]$? Often conjectured with $C = \tau \approx 1.176280818\ldots$, the Mahler measure of Lehmer's degree-10 Salem polynomial

$$L(x) = x^{10} + x^9 - x^7 - x^6 - x^5 - x^4 - x^3 + x + 1.$$

Status: OPEN. No peer-reviewed example with $1 < M < \tau$ is known (as of 2026-09 survey).

Box. `numpy.roots` + `mpmath` product (dps= 50) on 14 polynomials: $\Phi_1, \ldots, \Phi_6$ all return $M = 1$; $M(L) \approx 1.1762808182599200357$ (sanity OK); nearby Salem ≈ 1.20002652, ≈ 1.21639166; smallest Pisot $\theta_0 \approx 1.324717957$.

Problem statement

Lehmer conjecture / strong Salem form (OPEN). Is $\tau = M(L)$ the global minimum of $\{M(P) : P \in \mathbb{Z}[x],\ M(P) > 1\}$?

This experiment (Band-1). Literature ledger + reproducible box certificates for known anchors; **not** an exhaustive census; **not** a proof of minimality.

Literature ledger (not re-proved on box)

Year	Item	Authors	Status
1933	L; $M \approx 1.176280818$; gap question	D.H. Lehmer, *Ann. of Math.* (2) 34	classical champion
1971/79	Non-reciprocal: $M \geq \theta_0 \approx 1.3247\ldots$	C.J. Smyth	theorem
1977	Small Salem table (39)	D.W. Boyd, *Duke Math. J.* 44	published
1998+	Tables $M < 1.3$, deg$\leq$ 180–200; database	M.J. Mossinghoff, *Math. Comp.* 67; CECM / GitHub DB	published + DB
2006	Integer transfinite diameter; completeness class to deg 40	Rhin / Sac-Épée lineage	published
2024	11 further Salem in $(1.3, 49/37)$	J.-M. Sac-Épée, arXiv:2409.11159	published/in press
2025	Tiny measures $(1, 1.25]$; house/shortness; 236 known	El-Serafy–McKee, *Canad. Math. Bull.* 68	published
2026	Length-5 Salem complete (17/13); length-6 = 12 families + 126 sporadics (≥ 1.01)	McKee–Smyth, arXiv:2605.19013	preprint
2026	Claimed $M_{31} \approx 1.03878 < \tau$ (deg 162)	Helmdach, Zenodo 18874127	**non-canonical — NOT elevated**

Standing champion (peer-reviewed / database consensus): $\tau \approx 1.176280818\ldots$.

Method (box)

1. Fixed certificate list in `compute_mahler.py` (coeffs highest-degree first).
2. Roots via `numpy.roots` (float64); Mahler product via `mpmath` log-sum (`dps=50`).
3. Cyclotomic Φ_1–Φ_6 must give $M = 1$; Lehmer minimal + length-5 short form $z^{12} - z^7 - z^6 - z^5 + 1 = L \cdot \Phi_6$ must recover τ.
4. Nearby Salem / Pisot / trivial polys as regression checks.
5. **No claim** that the search space is exhaustive or that τ is proved minimal.

Results (numbers!)

Sanity / exclusions

Check	Result
Cyclotomic Φ_1–Φ_6	**all** $M = 1$ (OK)
$M(L)$ box	1.1762808182599200357
$\|M(L) - \tau_{\text{lit}}\|$	$< 10^{-10}$ (OK)
$M(L \cdot \Phi_6)$	1.1762808182599187035 (same τ)
Engine	`numpy.roots` + `mpmath` dps= 50
Wall	≈ 0.004 s

Certificate table (selected)

ID	Class	M (box)
Φ_1–Φ_6	cyclotomic	1
$x^2 - x - 1$	Pisot (golden)	1.6180339887498949025
$x^3 - x - 1$	smallest Pisot θ_0	1.3247179572447453921
Lehmer L	**Salem record**	1.1762808182599200357
Lehmer short len-5	Salem×cyclo	1.1762808182599187035
Salem σ_2 short	Salem	1.2000265239873906253
$z^{10}-z^6-z^5-z^4+1$	Salem	1.2163916611382672439
$x^2 - 2$	other	2
$x - 2$	Pisot	2

Full rows: `results.json`. Literature pointers: `literature_ledger.json`.

What this does not claim

- Lehmer's conjecture **solved** (it is **OPEN**).
- Existence / non-existence of $1 < M < \tau$.
- Completeness of Mossinghoff / El-Serafy–McKee censuses (beyond their stated hypotheses).
- Elevation of Zenodo Helmdach $M < \tau$ deposit to peer-reviewed champion status.

Next open (cited)

1. Global minimality of τ among $M > 1$ (Lehmer / strong Salem).
2. Complete tiny-measure census beyond shortness/house bounds of El-Serafy–McKee 2025.
3. Any example with $1 < M < \tau$ (**none known** in peer-reviewed literature).

Artifacts

- Code: `/workspace/experiments/band1/lehmer_mahler/compute_mahler.py`
- Results: `results.json`, `literature_ledger.json`, `SUMMARY.txt`
- Log: `/workspace/experiments/logs/2026-09-11_lehmer_mahler.log`
- Paper PDF: shelved in `DOWNLOAD_PDFs/`

References (anchors)

1. D.H. Lehmer, *Factorization of certain cyclotomic functions*, Ann. of Math. (2) **34** (1933), 461–479.
2. C.J. Smyth, *On the product of the conjugates outside the unit circle of an algebraic integer*, Bull. London Math. Soc. **3** (1971), 169–175.
3. M.J. Mossinghoff, *Polynomials with small Mahler measure*, Math. Comp. **67** (1998), 1697–1705; Mahler-Measure-Database.
4. S. El-Serafy, J. McKee, *Small Mahler measures with bounds on the house and shortness*, Canad. Math. Bull. **68** (2025), 603–619; DOI 10.4153/s0008439524000900.
5. J. McKee, C. Smyth, *Short Salem polynomials*, arXiv:2605.19013 (2026).